

IngeniumAI Hackathon – Non-contrast aorta & PA segmentation
Compiled 2026-08-13 16:02 UTC

Task: segment pulmonary artery and aorta on non-contrast chest CT
so PHAST can run earlier than contrast CTPA.

Method: contrast-removal domain adaptation (not plain supervised).
Source = unused contrast CTPA. Vessel HU is flattened to soft-tissue
range so SegResNet must use shape and location. Mixed with held-in
non-contrast cases under the same HU randomization.
Val = 3 held-out non-contrast cases (never trained on).

Device: mps
Non-contrast paired: 10 Contrast paired: 13
Epochs: 80 steps/epoch: 32
PA-focused sampling: pos=0.9 PA frac=0.7
Class weights (bg, PA, aorta): [0.4, 2.5, 1.0]

Mean hold-out Dice (non-contrast val)
Baseline PA 0.000 aorta 0.068 mean 0.034
Adapted PA 0.692 aorta 0.823 mean 0.758

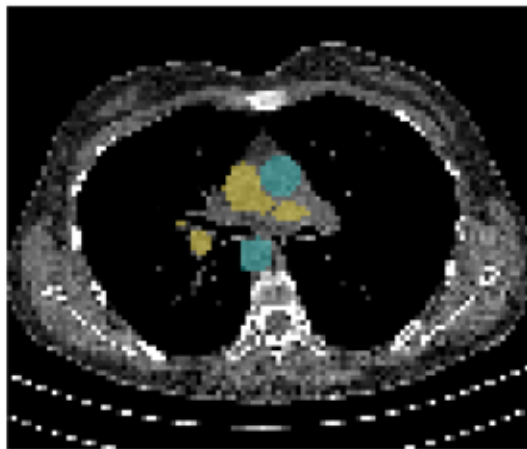
Clinical mean equivalent diameter at max-PA GT slice
GT MPA 56.9 mm aorta 42.6 mm ratio 1.317515847443455
Base MPA 0.0 mm aorta 13.9 mm ratio None
Adapt MPA 62.3 mm aorta 42.9 mm ratio 1.5043002840508601

Licence: CT-RATE is CC BY-NC-SA 4.0 (demonstration only).
A production PHAST model would be retrained on licensed/clinical data.
FUMPE / CAD-PE contrast sets are CC BY 4.0.

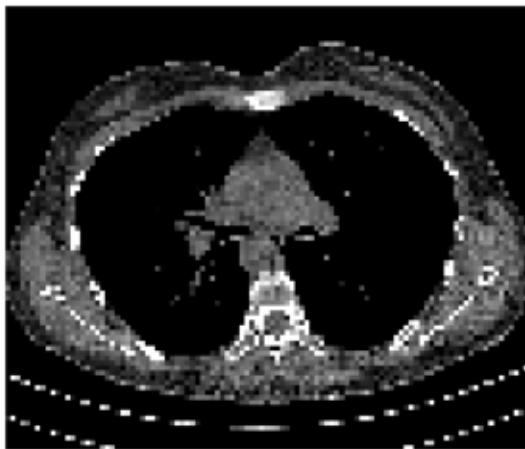
baseline train 10000 a 2.nii

baseline train_10000_a_2.nii PA Dice 0.00 aorta 0.00

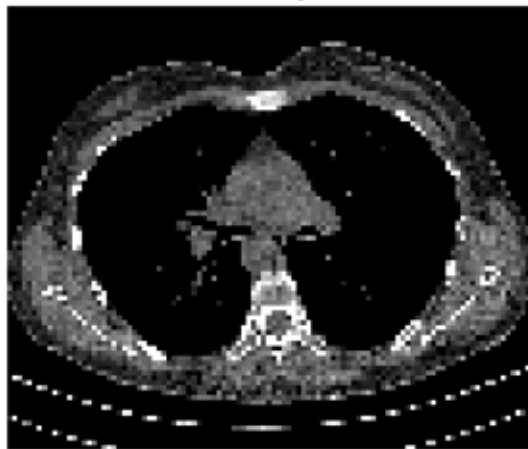
Ground truth



Prediction



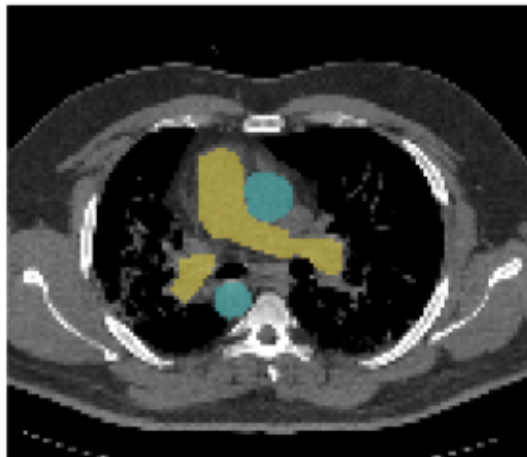
CT only



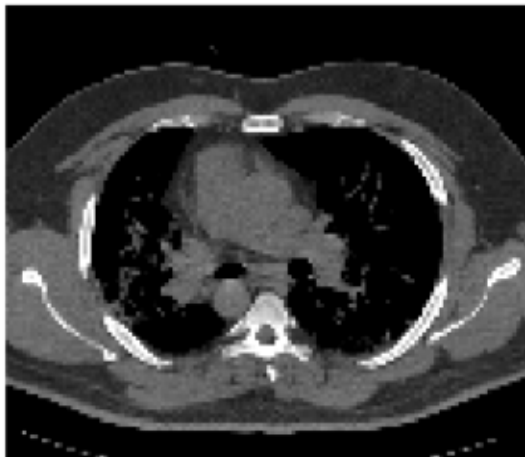
baseline train 10019 a 2.nii

baseline train_10019_a_2.nii PA Dice 0.00 aorta 0.02

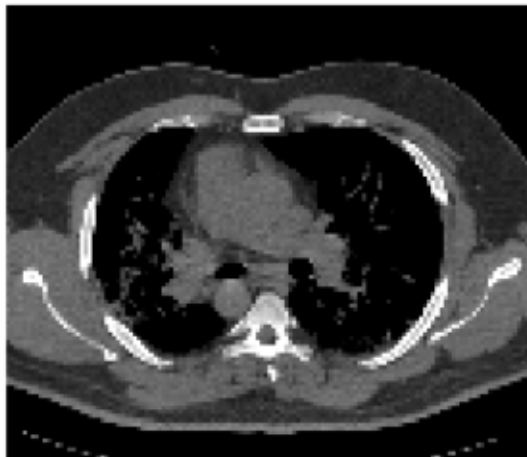
Ground truth



Prediction



CT only



baseline train 10020 a 2.nii

baseline train_10020_a_2.nii PA Dice 0.00 aorta 0.19

Ground truth



Prediction



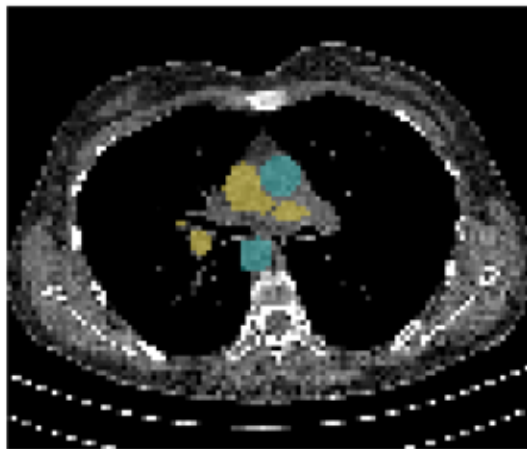
CT only



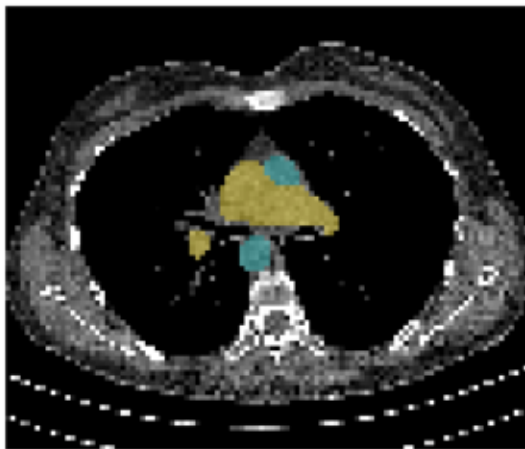
adapted train 10000 a 2.nii

adapted_train_10000_a_2.nii PA Dice 0.59 aorta 0.80

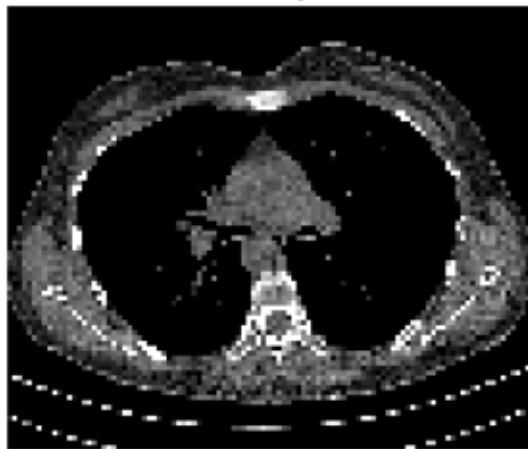
Ground truth



Prediction



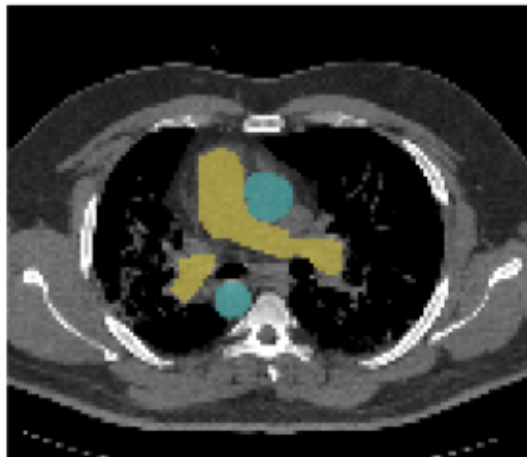
CT only



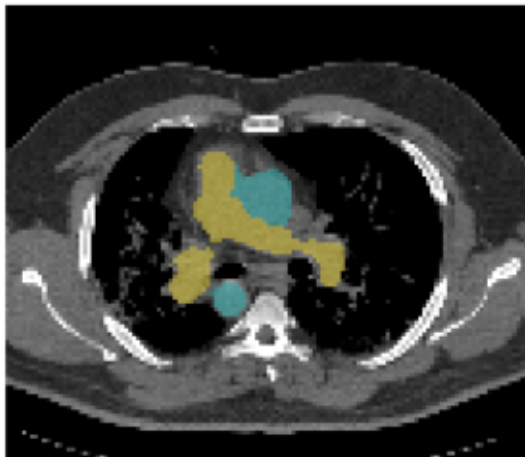
adapted train 10019 a 2.nii

adapted_train_10019_a_2.nii PA Dice 0.74 aorta 0.81

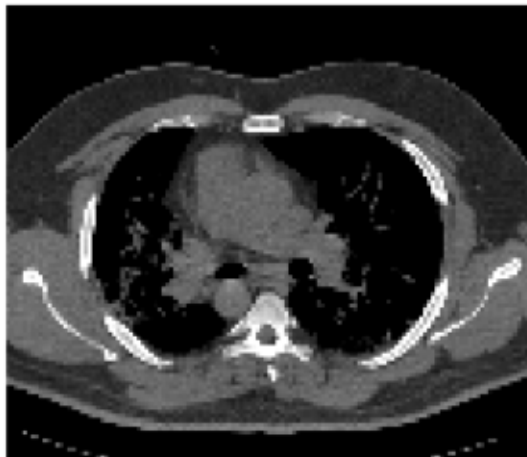
Ground truth



Prediction



CT only



adapted train 10020 a 2.nii

adapted_train_10020_a_2.nii PA Dice 0.75 aorta 0.86

Ground truth



Prediction



CT only

